

Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

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Code, checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

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Experimental Setting

Pipeline

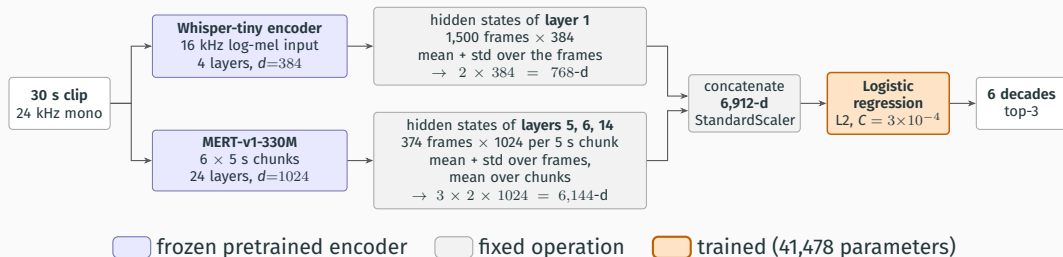
- 30 s clip, 24 kHz mono
- frozen encoder $\rightarrow$ mean + std pooling over time
- StandardScaler $\rightarrow$ L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

Hardware / software

GPU	2 $\times$ NVIDIA RTX A4500 (20 GB)
CPU / RAM	Intel i9-14900K, 125 GB
OS	Ubuntu 26.04
Software	PyTorch 2.14, Transformers 4.45, scikit-learn 1.9

Results

Task 1: model



- **Whisper-tiny, layer 1:** shallow layers work best for decade; accuracy drops with depth
- **MERT-v1-330M, layers 5, 6, 14:** best three of a 25-layer sweep; adding Whisper-tiny gives +2.3 points
- **mean + std pooling:** one d -dim vector per frame $\rightarrow$ mean and std **over the frames** $\rightarrow 2 \times d$ values per layer
- Whisper: **1,500 frames** per 30 s clip. MERT: **374 frames** per 5 s chunk (its training length); statistics are taken per chunk, then averaged over the six
- Only the classifier is trained, with strong L2 (1,026 clips, 6,912 features)

Task 1: release decade

Model: frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

Model and layers **chosen by ablation:** [▶ Section 3](#)

Validation, 132 clips	Top-1	Top-3
Best model	52.3%	86.4%

Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s (one neighbor each)

True	1960s	17	2	0	3	0	0
	1970s	5	9	3	2	2	1
	1980s	2	2	13	2	3	0
	1990s	1	3	2	9	4	3
	2000s	2	2	1	5	5	7
	2010s	0	2	0	0	4	16
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted					

Counts. Rows = true decade (22 clips each).

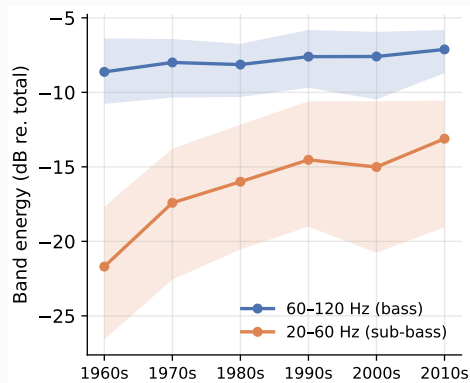
Why neighbors get confused (1): the low end

Hypothesis. Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

Analysis. Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

dB re. total	60s	70s	80s	90s	00s	10s
20-60 Hz	-21.7	-17.4	-16.0	-14.5	-15.0	-13.1
60-120 Hz	-8.6	-8.0	-8.1	-7.6	-7.6	-7.1

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

Why neighbors get confused (2): the loudness war

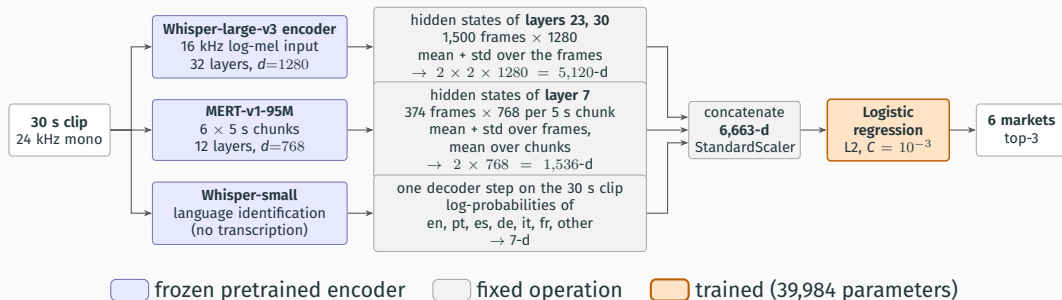
Hypothesis. From the 2000s, brick-wall limiters make masters louder and flatten peaks.

Median per decade, training clips	60s	70s	80s	90s	00s	10s
RMS level (dBFS)	-14.0	-14.6	-14.5	-13.0	-11.0	-10.6
Crest factor, peak – RMS (dB)	13.9	14.5	14.5	13.0	11.0	10.6
Dynamic range, loudest 20% blocks (dB)	10.6	11.2	11.3	10.7	8.9	8.7
Longest run pinned at the peak (samples)	5	5	6	9	22	26
Samples within 0.5 dB of the peak (%)	0.02	0.01	0.02	0.06	0.42	0.50

- 2000s and 2010s clips are 3 dB louder, have 3 dB less crest, and hold peaks at the ceiling. The two are **almost indistinguishable**, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

Caveat. The clips are 24 kHz mono, decoded from lossy Opus (YouTube). Resampling, downmixing and lossy coding reshape waveform peaks, so these numbers are trends (ANOVA $p < 10^{-30}$), not exact properties of the original masters. The two sample-level rows are the most affected.

Task 2: model



- **Whisper-large-v3, layers 23, 30**: deepest layers work best, the opposite of Task 1; this fits language being the main cue
- **MERT-v1-95M, layer 7**: +2 points on top of Whisper (57.8% $\rightarrow$ 59.8%); a music encoder adds what a speech encoder misses
- **Whisper-small language ID**: an explicit 7-d posterior; a small gain (top-3 89.2% $\rightarrow$ 90.2%) because deep Whisper layers already encode language
- Pooling as in Task 1: 1,500 Whisper frames per clip; 374 MERT frames per 5 s chunk

Task 2: release market

Model: frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

Model and layers **chosen by ablation:** [▶ Section 3](#)

Validation, 102 clips	Top-1	Top-3
Best model	60.8%	90.2%

Errors

- **US ↔ UK** is the largest confusion (9 clips)
- **UK is over-predicted:** 23 called UK, only 9 correct
- Brazil (.88) and Spain (.82) are rarely missed
- Germany, Italy: 12 of their 21 errors go to US / UK

True	US	11	4	0	0	1	1
	UK	5	9	0	0	2	1
	Brazil	0	0	15	0	0	2
	Spain	0	2	0	14	1	0
	Germany	3	4	1	2	6	1
	Italy	1	4	0	2	3	7
		US	UK	Brazil	Spain	Germany	Italy
		Predicted					

Counts. Rows = true market (17 clips each).

Why English-language markets get confused: language

The dataset has **no language label**. We estimate it with Whisper-small **language identification** on the 798 training clips, on the mixture and on Demucs-separated vocal stems.

Market	Home language	Home mixture	Detected as English	
			mixture	vocals
US	English	92%	–	–
UK	English	100%	–	–
Brazil	Portuguese	87%	11%	12%
Spain	Spanish	57%	39%	41%
Germany	German	17%	83%	82%
Italy	Italian	41%	55%	56%

Share of training clips whose most likely language (among en, pt, es, de, it) is the home language or English.

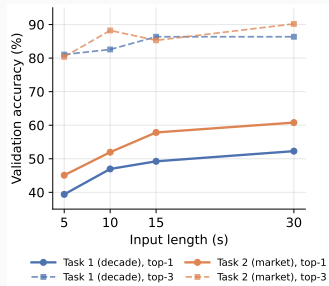
- Most German releases and over half of the Italian ones are **sung in English**, so their vocals carry no market cue
- Vocal stems give the same answer: the accompaniment is not what misleads the detector
- Language separates Brazil and Spain. The English-language group has to be split by genre and production style, a much weaker cue
- A release market is where the record was issued, not the artist's nationality or language

Optional Experiments

Input length: 5 / 10 / 15 / 30 s

Best models on the first k seconds only; features re-extracted and classifier re-trained per length.

Length	Task 1		Task 2	
	Top-1	Top-3	Top-1	Top-3
5 s	39.4	81.1	45.1	80.4
10 s	47.0	82.6	52.0	88.2
15 s	49.2	86.4	57.8	85.3
30 s	52.3	86.4	60.8	90.2



- Top-1 rises monotonically on both tasks and has **not saturated** at 30 s; Task 1 top-3 is flat after 15 s
- Task 2 gains more (+15.7 vs. +12.9 points)

Multiple excerpts per recording

Each 30 s clip is cut into six 5 s chunks. **Chunk-level:** every chunk is its own training sample ($6\times$ more data) and chunk probabilities are averaged at test time. **Clip-level:** chunk features are averaged first (our default).

Task	Features	Clip-level		Chunk-level	
		Top-1	Top-3	Top-1	Top-3
1	MERT-v1-95M, layers 6, 8, 10	48.5	81.8	47.7	84.8
1	MERT-v1-330M, layers 5, 6, 14	50.0	87.9	50.0	86.4
2	MERT-v1-95M, layer 7	42.2	72.5	45.1	76.5
2	MERT-v1-330M, layers 8, 9, 12	42.2	69.6	44.1	76.5

- Task 1: no gain. Six chunks of one recording are highly correlated, so they add little new information for a linear model
- Task 2: a small gain (+2 to +3 points top-1, +4 to +7 top-3), about 2 to 3 validation clips
- We keep clip-level features for the final models: same accuracy, six times fewer training rows

MERT features only, not the full final models: Whisper features were extracted on whole 30 s clips.

Task 1 as year regression

Decades are ordered, so we test a formulation that uses the order.

Features (same as the final Task 1 model):

Whisper-tiny encoder layer 1 + MERT-v1-330M layers 5, 6, 14, mean + std pooling, 6,912-d.

Classification: L2 logistic regression, $C = 3 \times 10^{-4}$.

Regression: ridge regression, $\alpha = 10^4$, on a decade index (0 = 1960s, 1 = 1970s, ..., 5 = 2010s), rounded to the nearest decade; top-3 = the three nearest decades.

Formulation	Top-1	Top-3
Logistic regression (final model)	52.3	86.4
Ridge regression	31.8	77.3

Mean absolute error of the regression: 0.98 decades.

True	1960s	3	10	6	3	0	0
	1970s	1	8	8	4	1	0
	1980s	0	2	10	9	1	0
	1990s	0	0	5	9	7	1
	2000s	0	1	5	6	9	1
	2010s	0	1	1	7	10	3
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted					

Ridge regression. Counts; rows = true decade (22 clips each).

Task 1 as year regression: why it fails

Both rows use the final Task 1 features (Whisper-tiny layer 1 + MERT-v1-330M layers 5, 6, 14); only the predictor differs.

Number of validation clips each formulation assigns to each decade:

Clips predicted as	1960s	1970s	1980s	1990s	2000s	2010s
True count	22	22	22	22	22	22
Logistic regression (final model)	27	20	19	21	18	27
Ridge regression	4	22	35	38	28	5

- **Regression is much worse** (−20.5 points top-1): predictions collapse toward the middle, only 4 clips are called 1960s and 5 are called 2010s
- Tuning does not fix it: with a ridge penalty from 10^2 to 10^6 the best top-1 is 32%; stronger regularization worsens the collapse
- The classifier already learns the order by itself (see the t-SNE slide), so nothing is gained from imposing it

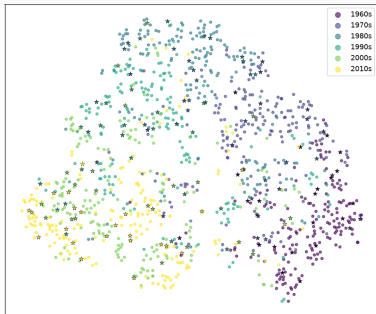
Task 2: mixture vs. vocals vs. accompaniment

Demucs (htdemucs) separates each clip into vocals and accompaniment. MERT-v1-95M layer 7 features are extracted from each input.

Input to MERT	Top-1	Top-3
Mixture	42.2	72.5
Vocals	43.1	78.4
Accompaniment	33.3	61.8
Mixture + vocals + accompaniment	46.1	76.5
Mixture + language ID	55.9	77.5
Vocals + language ID	55.9	79.4
Accompaniment + language ID	47.1	74.5

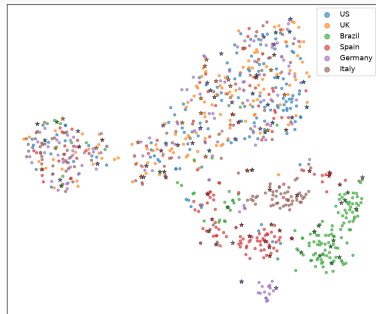
- **Market information is in the vocals:**
removing them costs 9 points, keeping only them costs nothing
- Vocals halve the US $\leftrightarrow$ UK confusions (9 $\rightarrow$ 4 clips). The accompaniment does not help (9 clips), and its US recall is 18%
- With language ID added, vocals and mixture tie at 55.9%, so the final model uses the mixture: no separation needed at inference

t-SNE of the best models



Task 1: 6-d logit space of the classifier

- Task 1: decades line up along **one arc from the 1960s to the 2010s**, although the classifier was never told they are ordered; 2000s and 2010s overlap
- Task 2: Brazil, Spain and Italy form clusters **before any training**; US, UK and most of Germany share one English-language cloud



Task 2: raw feature space, no classifier

Dots = train, stars = validation. Train points in the logit space are fitted; only the stars show generalization.

Confusion patterns and dataset bias

The detailed analyses are in the Results section. Summary, plus two additional findings:

	Task 1: decade	Task 2: market
Confusion pattern	57% of errors on a neighboring decade; 2000s ↔ 2010s is the worst pair	75% of errors among US, UK, Germany, Italy; UK is over-predicted
Bias in the data	production changes with the era: sub-bass +8.6 dB, crest factor −3 dB after 2000	English-language releases: 83% of Germany, 55% of Italy
What the model keys on	mastering loudness and bandwidth as well as musical content	the language of the vocals rather than the market itself

- **Hand-crafted production features** (24-d: band energies, crest factor, limiter statistics) reach 40.9% on Task 1 but only **21.6% on Task 2** (chance 16.7%): within one decade, mastering does not depend on the market
- **High-frequency energy** (8–12 kHz) is 4 to 5 dB lower in the 1960s and 1970s than later; the clips are band-limited to 12 kHz, so anything above that is unavailable

Ablation

Frozen features vs. fine-tuning

Audio Language Model

Disclosure

References
